

UYEN HUYNH

Portfolio [◇ GitHub](#)

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PROFESSIONAL SUMMARY

AI / ML Engineer with 2+ years of experience in healthcare AI, applied machine learning, and research-driven software projects. Built ML pipelines, inference APIs, and data workflows using Python, FastAPI, Docker, PostgreSQL, Redis, PyTorch, and LightGBM. Background in biotechnology with strong interest in healthcare AI, bioinformatics, and practical ML systems.

TECHNICAL SKILLS

Languages & Data: Python, SQL, PostgreSQL

ML / DL: PyTorch, TensorFlow, LightGBM, Scikit-learn, YOLOv8, SHAP

Backend & Deployment: FastAPI, REST APIs, Docker, Redis, GitHub Actions, CI/CD

LLM Applications: LangChain, RAG, Qdrant, Chroma, semantic search

Tools: Git, GitHub, Pytest

Domain Knowledge: healthcare data handling, PII anonymization, API security basics

WORK EXPERIENCE

AI Research Collaborator

May 2024 – Present

Taipei Medical University, Taipei City, Taiwan

- Built and evaluated machine learning pipelines for drug interaction and drug toxicity prediction using molecular and structured biomedical data.
- Contributed to experiment design, model analysis, and manuscript preparation in a collaborative healthcare AI research environment.

AI Researcher / Teaching Assistant

Jan 2023 – Present

AI Vietnam, Ho Chi Minh City, Vietnam - Part time, Contract

- Contributed to LLM, NLP, and healthcare-focused AI projects through data preparation, experimentation, and evaluation workflows.
- Built Python-based preprocessing and analysis pipelines for research and educational projects, and supported implementation debugging in project-based settings.

Bioinformatics Research / Teaching Assistant

Aug 2024 – May 2025

Thong Nhat Hospital, Ho Chi Minh City, Vietnam

- Supported a breast cancer classification project through biomedical data preprocessing, feature preparation, and model evaluation.
- Assisted with foundational bioinformatics instruction and practical data analysis activities in a clinical and research-oriented setting.

Software Development Intern

Jul 2024 – Sep 2024

Mat Bao Corporation, Ho Chi Minh City, Vietnam

- Supported backend development for internal web applications using .NET and SQL.
- Worked on feature updates, bug fixes, and database-related tasks in a team environment.

SELECTED PROJECTS

MediAI – ICU Risk Prediction Platform [\[GitHub\]](#) [\[Demo\]](#)

Personal Project — FastAPI, LightGBM, Redis, Docker, PostgreSQL, Next.js

- Built an end-to-end machine learning application for ICU risk prediction using 70,000+ MIMIC-IV ICU stays.
- Developed a FastAPI inference service with Redis caching and PostgreSQL storage for prediction results.
- Reduced single-request inference latency to around 50 ms in local deployment tests and integrated a lightweight RAG chatbot for medical queries.

T-DDI – Drug Interaction Prediction [\[Live Demo\]](#)

Team Research Project — PyTorch, TabTransformer, QSAR Descriptors, SMILES, LIME

- Developed a drug-drug interaction prediction framework evaluated on 878k+ DrugBank pairs.
- Built a feature engineering pipeline that converted SMILES into 3,780 molecular descriptors for structured model input.
- Achieved competitive results against internal baseline models and applied LIME-based interpretation to examine important chemical features.

DeepGPT-DILI – Drug-Induced Liver Injury Prediction [\[GitHub\]](#) [\[Publication\]](#)

Team Research Project — PyTorch, GCN, XGBoost, SMILES

- Contributed to an ensemble framework for drug toxicity prediction on the DILIRank dataset.
- Built preprocessing and feature engineering workflows for molecular structures and SMILES-based inputs.
- Contributed to work associated with a publication accepted at MICCAI 2025.

Breast Cancer Classification Model

Bioinformatics / Healthcare ML Project

- Developed a machine learning model for classifying benign and malignant breast cancer cases from biomedical features.
- Compared model behavior through structured evaluation and error analysis to improve interpretability.

EDUCATION

Bachelor of Engineering in Biotechnology

Expected 2026

University of Agriculture and Forestry, Ho Chi Minh City, Vietnam

ADDITIONAL INFORMATION

Peer Reviewer: PLOS ONE

Languages: Vietnamese (Native), English (B2), French (B2), Chinese (HSK 4)